

Assignment-1

Question	RBT Level	Course Outcome
1. a) An electricity board charges the following rates to domestic users to discourage large consumption of energy: For the first 100 units- 60P per unit For next 200 units- 80P per unit Beyond 300 units- 90P per unit All users are charged a minimum of Rs. 50.00. if the total amount is more than Rs. 300.00 then an additional surcharge of 15% is added. Build a program using object oriented approach to read the names of users and number of units consumed and print out the charges with names. b) Create two classes A and B which store the value of distances. 'A' stores distances in meters and centimeters and 'B' in feet and inches. Construct a program that can add these distances of class A and class B, and display the result in the format of meters and centimeters. Use a friend function to carry out the addition operation.	L6	CO1, CO2, CO6
2. a) A book shop maintains the inventory of books that are being sold at the shop. The list includes details such as author, title, price, publisher and stock position. Whenever a customer wants a book, the sales person inputs the title and author and the system searches the list and displays whether it is available or not. If it is not, an appropriate message is displayed. If it is, then the system displays the book details and requests for the number of copies required. If the requested copies are available, the total cost of the requested copies is displayed; otherwise, the message "Required copies not in stock" is displayed. Design a system using a class 'books' with suitable member functions and constructors. Use 'new' operator in constructors to allocate memory space required. b) An educational institution wishes to maintain a database of its employees. The database is divided into a number of classes as shown in figure. Specify all the classes and define functions to create the database and retrieve individual information as and when required.	L6	CO1, CO2, CO3, CO6

